

The Institutional Stance (Jara-Ettinger & Dunham, 2024)

Prepared by Joseph Low

Research Engineer @ Metagov

October 22, 2025



Metagov



Public AI



Outline

1. Introduction
2. Three Stages of the Institutional Stance
3. Institutions as Graphs
4. Interplay with Mentalizing
5. Conclusion

Introduction



The Puzzle: Friday at the Office



Imagine you're interning at a company.

The Puzzle: Friday at the Office



Imagine you're interning at a company.

Your boss comes in and asks you:
“Can you help me get lunch and print out these documents?”

The Puzzle: Friday at the Office



Imagine you're interning at a company.

Your boss comes in and asks you:
“Can you help me get lunch and print out these documents?”

You get it done.

The Puzzle: Saturday at the Movies



You go to watch a movie.

The Puzzle: Saturday at the Movies



You go to watch a movie.

Then your boss spots you from a few rows ahead and asks you to get popcorn for them.

The Puzzle: Saturday at the Movies



You go to watch a movie.

Then your boss spots you from a few rows ahead and asks you to get popcorn for them.

Would you do this?

The Mentalistic Stance Prediction

At the office:

Boss desires: lunch, documents

Boss believes: intern will get it if
asked

The Mentalistic Stance Prediction

At the office:

Boss desires: lunch, documents

Boss believes: intern will get it if asked

Intern desires: to help boss

Intern believes: they can fulfill this task

The Mentalistic Stance Prediction

At the office:

Boss desires: lunch, documents

Boss believes: intern will get it if asked

Intern desires: to help boss

Intern believes: they can fulfill this task

At the movies:

Boss desires: popcorn

Boss believes: intern will get it if asked

The Mentalistic Stance Prediction

At the office:

Boss desires: lunch, documents

Boss believes: intern will get it if asked

Intern desires: to help boss

Intern believes: they can fulfill this task

At the movies:

Boss desires: popcorn

Boss believes: intern will get it if asked

Intern desires: to help boss

Intern believes: they can fulfill this task

The Mentalistic Stance Prediction

At the office:

Boss desires: lunch, documents

Boss believes: intern will get it if asked

Intern desires: to help boss

Intern believes: they can fulfill this task

Theory of Mind says: Movies scenario should be unsurprising

At the movies:

Boss desires: popcorn

Boss believes: intern will get it if asked

Intern desires: to help boss

Intern believes: they can fulfill this task

The Mentalistic Stance Prediction

At the office:

Boss desires: lunch, documents

Boss believes: intern will get it if asked

Intern desires: to help boss

Intern believes: they can fulfill this task

At the movies:

Boss desires: popcorn

Boss believes: intern will get it if asked

Intern desires: to help boss

Intern believes: they can fulfill this task

Theory of Mind says: Movies scenario should be unsurprising

- Same mental states, same belief-desire structure

The Mentalistic Stance Prediction

At the office:

Boss desires: lunch, documents

Boss believes: intern will get it if asked

Intern desires: to help boss

Intern believes: they can fulfill this task

At the movies:

Boss desires: popcorn

Boss believes: intern will get it if asked

Intern desires: to help boss

Intern believes: they can fulfill this task

Theory of Mind says: Movies scenario should be unsurprising

- Same mental states, same belief-desire structure

But it feels WRONG. Why?

The Institutional Stance Explanation

At the office:

Boss occupies role: “CEO”

Intern occupies role: “assistant”

The Institutional Stance Explanation

At the office:

Boss occupies role: “CEO”

Intern occupies role: “assistant”

Role-based norms: CEO can
delegate tasks, assistant complies

The Institutional Stance Explanation

At the office:

Boss occupies role: “CEO”

Intern occupies role: “assistant”

Role-based norms: CEO can
delegate tasks, assistant complies

The Institutional Stance Explanation

At the office:

Boss occupies role: “CEO”

Intern occupies role: “assistant”

Role-based norms: CEO can
delegate tasks, assistant complies

At the movies:

Boss: person at a movie theater

Intern: person on a date

The Institutional Stance Explanation

At the office:

Boss occupies role: “CEO”

Intern occupies role: “assistant”

Role-based norms: CEO can delegate tasks, assistant complies

At the movies:

Boss: person at a movie theater

Intern: person on a date

CEO/assistant roles are NOT ACTIVE in this context

The Institutional Stance Explanation

At the office:

Boss occupies role: “CEO”

Intern occupies role: “assistant”

Role-based norms: CEO can delegate tasks, assistant complies

At the movies:

Boss: person at a movie theater

Intern: person on a date

CEO/assistant roles are NOT ACTIVE in this context

Behavior violates norms

The Institutional Stance Explanation

At the office:

Boss occupies role: “CEO”

Intern occupies role: “assistant”

Role-based norms: CEO can delegate tasks, assistant complies

Humans automatically track which roles are operative in each context

At the movies:

Boss: person at a movie theater

Intern: person on a date

CEO/assistant roles are NOT ACTIVE in this context

Behavior violates norms

The Rental Car Example

Laurie arrives at a rental car agency. Paul greets her and helps with the paperwork.

The Rental Car Example

Laurie arrives at a rental car agency. Paul greets her and helps with the paperwork.

Fee is higher than expected → argument starts.

The Rental Car Example

Laurie arrives at a rental car agency. Paul greets her and helps with the paperwork.

Fee is higher than expected → argument starts.

Paul briefly excuses himself. A few minutes later, Alex appears and picks up the discussion with Laurie **right where Paul left off.**

The Rental Car Example

Laurie arrives at a rental car agency. Paul greets her and helps with the paperwork.

Fee is higher than expected → argument starts.

Paul briefly excuses himself. A few minutes later, Alex appears and picks up the discussion with Laurie **right where Paul left off**.

Mentalistic stance: Transfer of Paul's beliefs/desires to Alex is strikingly odd!

Institutional stance: Unsurprising. Same role = same goals/behaviors.

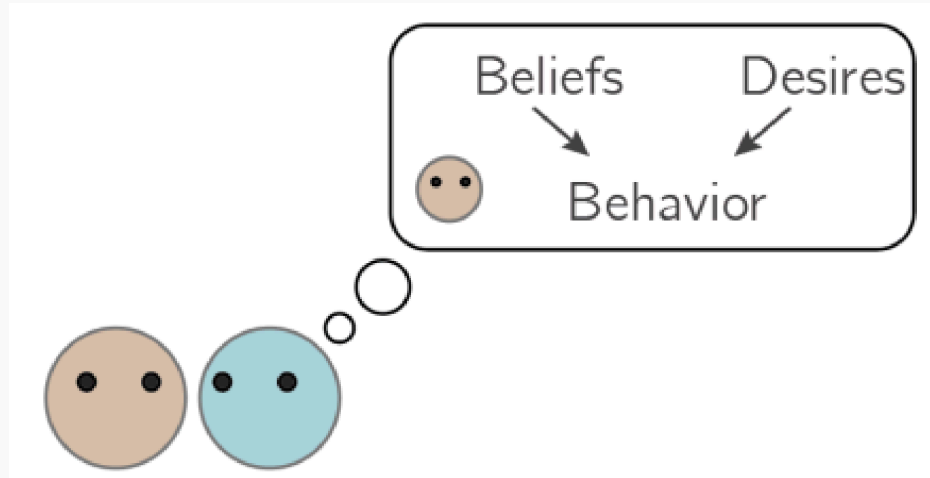
Compare to Personal Relationships

Imagine during a personal argument with a close friend, the friend excuses themselves and a casual acquaintance steps in to continue arguing...

Now it would be bizarre!

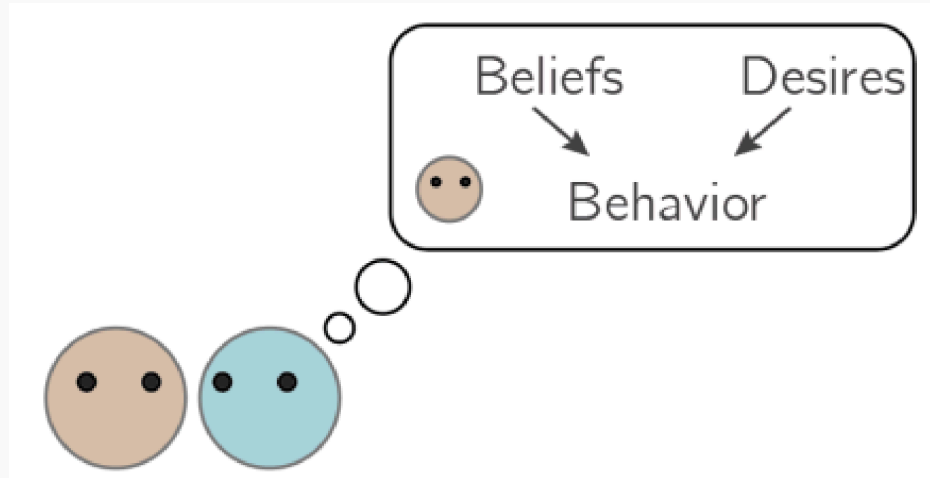
The institutional stance is **not** warranted in personal relationships—transfer of goals would be surprising, if not off-putting.

Mentalistic Stance vs Institutional Stance visualized

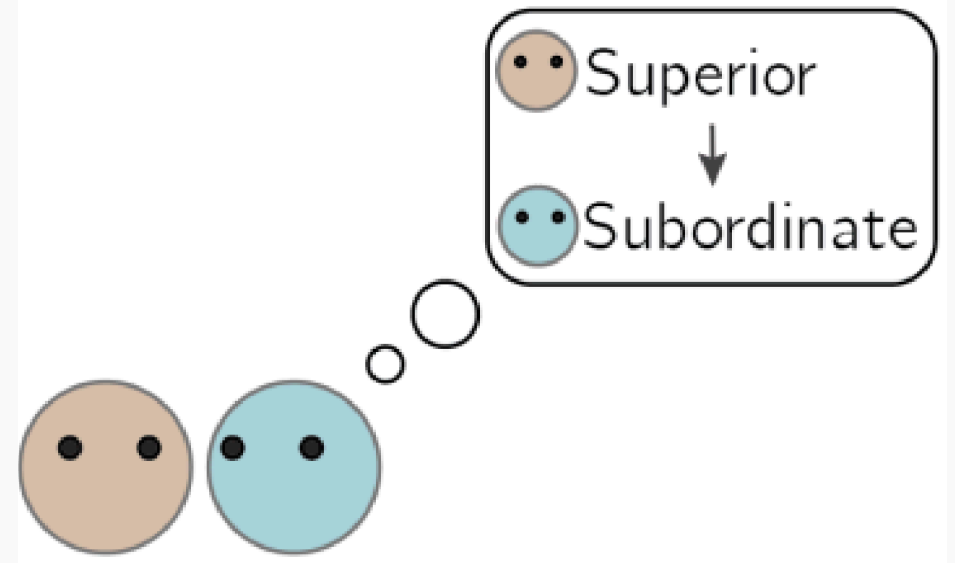


Mentalistic Stance: Individual beliefs and desires

Mentalistic Stance vs Institutional Stance visualized



Mentalistic Stance: Individual beliefs and desires



Institutional Stance: Roles and normative relations

Three Stages of the Institutional Stance

The Puzzle of Animal Social Coordination

Theory of Mind is powerful but cannot explain all social complexity

The Puzzle of Animal Social Coordination

Theory of Mind is powerful but cannot explain all social complexity

Many species exhibit complex social coordination and cooperation

The Puzzle of Animal Social Coordination

Theory of Mind is powerful but cannot explain all social complexity

Many species exhibit complex social coordination and cooperation

Yet these species lack the sophisticated mentalizing abilities of humans

The Puzzle of Animal Social Coordination

Theory of Mind is powerful but cannot explain all social complexity

Many species exhibit complex social coordination and cooperation

Yet these species lack the sophisticated mentalizing abilities of humans

The Question: If animals don't have Theory of Mind, how do they coordinate and cooperate with one another?

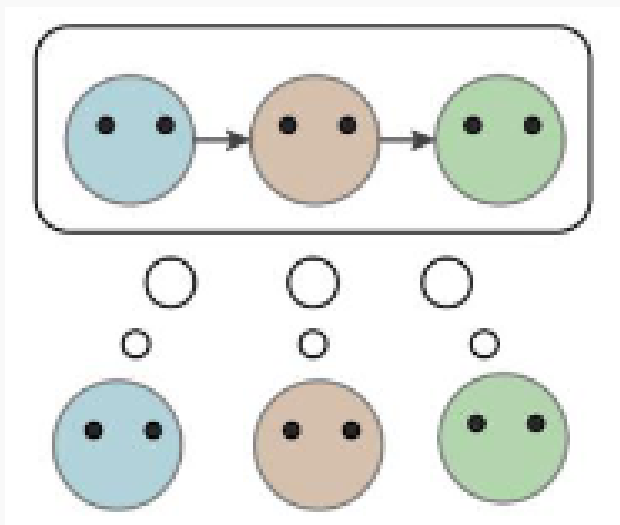
A Hypothesis: Three Critical Stages

Solutions to social coordination evolved through three critical stages:

1. **Proto-institutional:** Rigid role-based systems (biologically determined)
2. **Fixed institutional:** Role-filler independence (roles can be swapped)
3. **Generative institutional:** Grammar of institutions (uniquely human)

Each stage adds flexibility and representational power

Stage 1: Proto-Institutional Representations

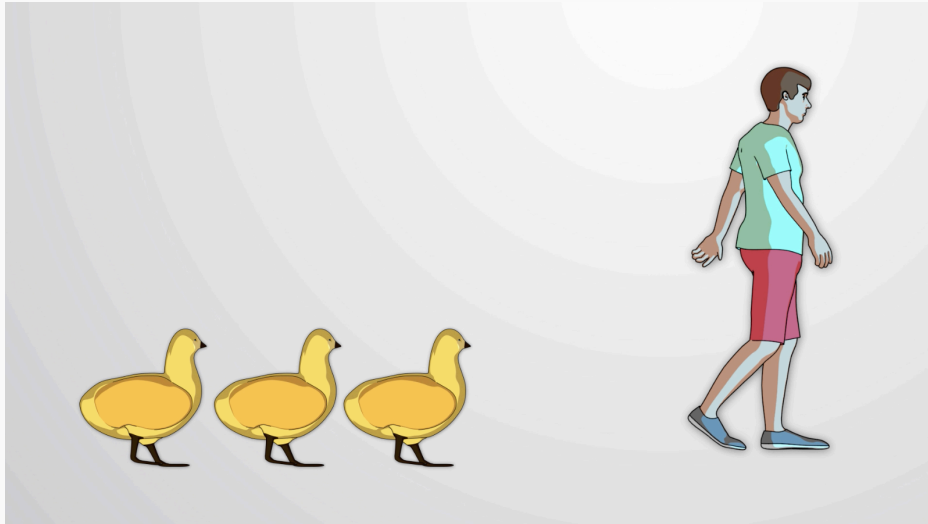


The core idea: Learn that specific individuals occupy specific roles

This allows us to:

- Predict how this individual will act
- Understand how to interact with them
- Different roles = different social affordances

Stage 1 Example: Imprinting



Shortly after hatching, ducklings identify the first moving object as “mother”

Proto-institutional representation:

- Learn to identify who occupies the “mother” role
- Associate this specific individual with caregiving behaviors
- To the duckling, mother-role and occupier of role are fused

Stage 1: Key Limitation

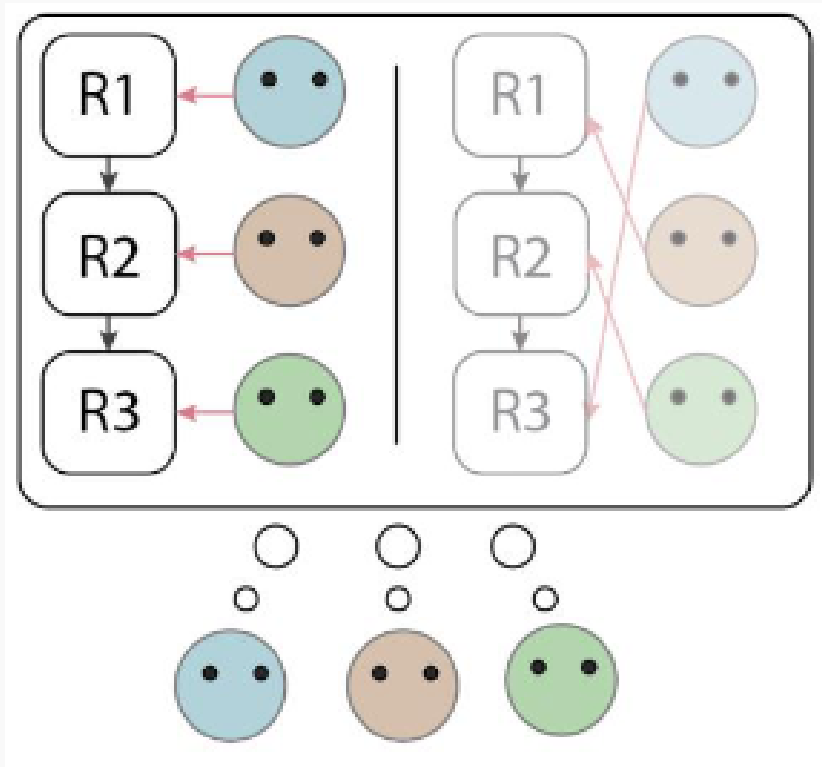
No abstract representation of role vs. occupier

The agent IS the role—just an association of specific individuals to social affordances

No role swapping possible

Still enables coordination without mentalizing

Stage 2: Fixed Institutional Representations



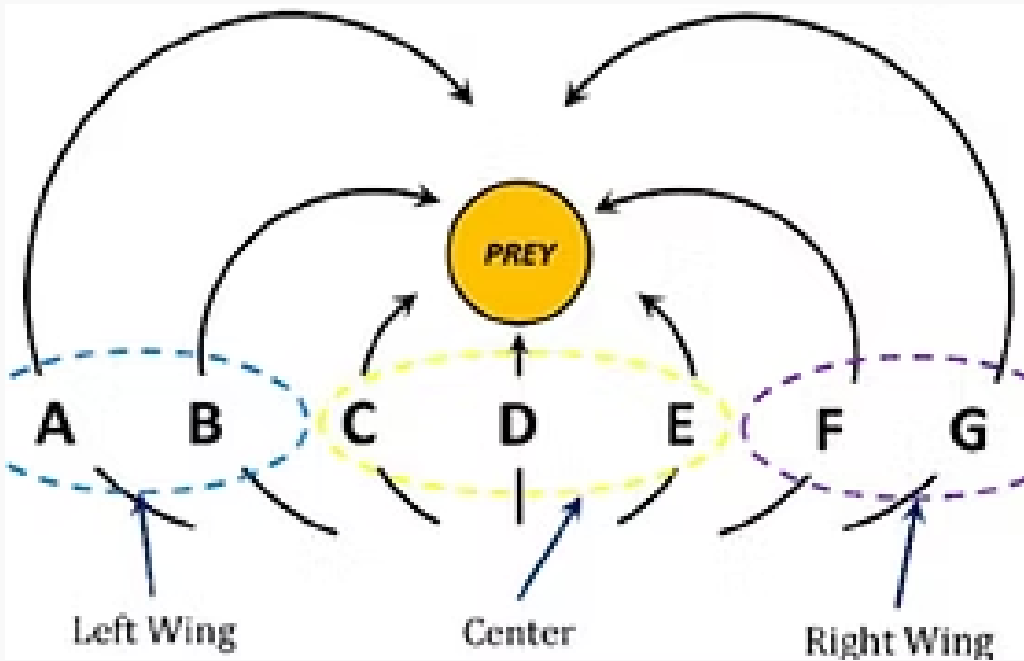
The core idea: Any individual can occupy specific roles

Abstract representation of roles independent of who occupies them

Agents can track and remember:

- Which role an individual occupies
- When roles change
- How to adjust behavior accordingly

Stage 2 Example: Lions Coordinated Hunting



Lions hunt in coordinated formations:
center hunters and wing hunters

Fixed institutional representation:

- Role specialization with ability to switch positions
- Must understand all roles need filling
- Know which role each lion currently occupies
- Can adapt when roles change

Stage 2: Key Limitation

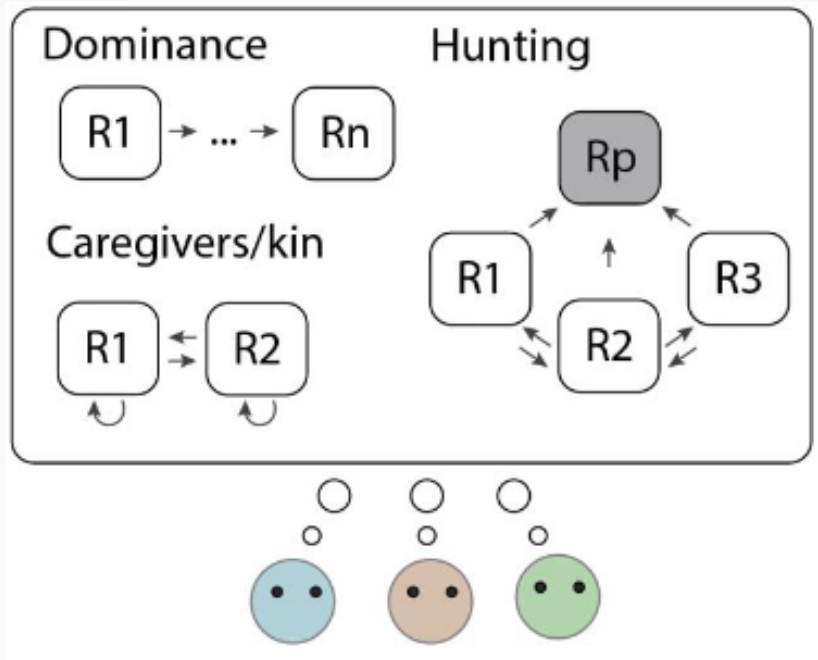
The set of roles available to interpret behavior are pre-determined

No cultural evolution or emergence of novel institutional structures

Cannot create new roles or institutional arrangements

The institutional representations are fixed—determined by biology or evolutionary history

Stage 3: Generative Institutional Representations



The core idea: Generative system to synthesize novel institutional structures

This enables us to:

1. Identify normatively structured systems of interconnected roles
2. Compositionally construct new such systems
3. Use that knowledge to interact with others

Stage 3 Example: Attending Catholic Mass as Non-Catholic



Even without prior experience, you can understand mass as an institutional structure:

- Identify roles: priest, altar servers, congregation
- Recognize hierarchical relations and ritual patterns
- Compositionally combine: religious authority + ceremony + worship
- Navigate appropriately without explicit instruction

Institutions as Graphs

What is an Institution?

An institution can be modelled as a graph of roles with normative relations

What is an Institution?

An institution can be modelled as a graph of roles with normative relations

Nodes = Roles

- Purpose: Task and Function. Who can occupy the role.
- Power: Powers granted by the role
- Limitations: Obligations imposed by it

What is an Institution?

An institution can be modelled as a graph of roles with normative relations

Nodes = Roles

- Purpose: Task and Function. Who can occupy the role.
- Power: Powers granted by the role
- Limitations: Obligations imposed by it

Edges = Relations between roles

- How roles interact with one another
- Can be directional or undirectional
- Can grant social power over another role

What is an Institution?

An institution can be modelled as a **graph of roles with normative relations**

Nodes = Roles

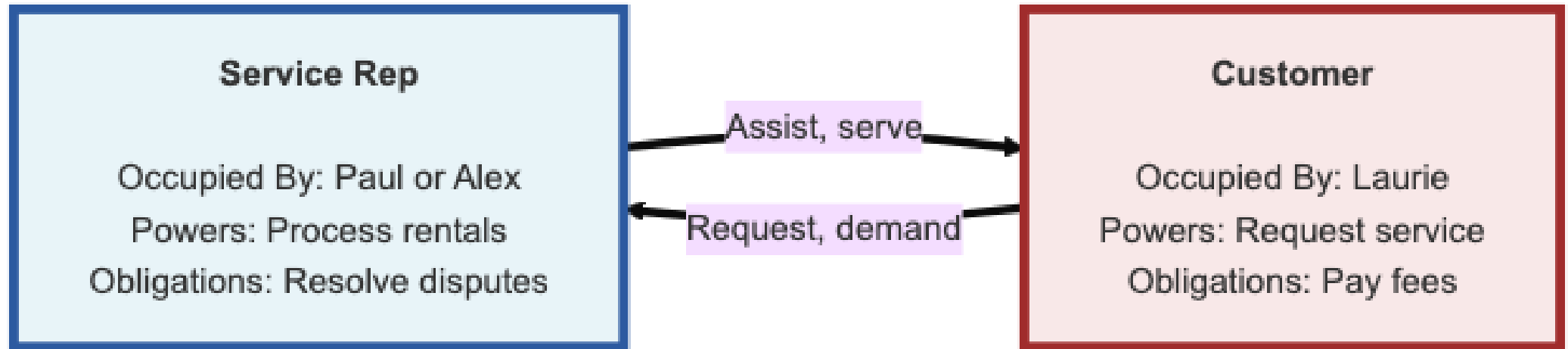
- Purpose: Task and Function. Who can occupy the role.
- Power: Powers granted by the role
- Limitations: Obligations imposed by it

Edges = Relations between roles

- How roles interact with one another
- Can be directional or undirectional
- Can grant social power over another role

Key insight: Roles are not isolated—they're understood as part of a broader institutional structure

Example: Rental Car Agency



Role-Filler Independence: Paul and Alex can both occupy the Service Rep role. When Alex replaces Paul, Alex inherits all norms of the role.

Discussion Question #1

- Do people understand each other in ways that cannot be reduced to mentalizing?
 - What does it mean to reduce social reasoning to mentalizing?

Social Ontology: What is the nature of the social world?

Two philosophical stances in social ontology:

Methodological Individualism

The view that social phenomena can and should be reduced to individual agents, their beliefs, actions, and interactions.

All social facts are ultimately facts about individuals.

Social Ontology: What is the nature of the social world?

Two philosophical stances in social ontology:

Methodological Individualism

The view that social phenomena can and should be reduced to individual agents, their beliefs, actions, and interactions.

All social facts are ultimately facts about individuals.

Methodological Collectivism

The view that social phenomena have irreducible collective properties that cannot be fully reduced to individual-level facts.

Social groups and institutions have properties and causal powers of their own.

Discussion Question #1

- Do people understand each other in ways that cannot be reduced to mentalizing?
 - What does it mean to reduce social reasoning to mentalizing?

Is mentalizing an individual or collective action?

The Institutional Stance as Non-Reductive Theory

Relates to work that straddles methodological individualism/collectivism

1. Representing Collectives

Categories like race and
gender as structural
positions rather than
essential properties

The Institutional Stance as Non-Reductive Theory

Relates to work that straddles methodological individualism/collectivism

1. Representing Collectives

Categories like race and gender as structural positions rather than essential properties

2. Representing Socially-Structured Behavior

Norms enforced through social pressure (injunctive) or that simply emerge (descriptive)

The Institutional Stance as Non-Reductive Theory

Relates to work that straddles methodological individualism/collectivism

1. Representing Collectives

Categories like race and gender as structural positions rather than essential properties

2. Representing Socially-Structured Behavior

Norms enforced through social pressure (injunctive) or that simply emerge (descriptive)

3. Representing Interactions and Scripts

Schemas and scripts we learn, e.g. how to order coffee in a fancy cafe

Discussion Question #2

Are institutions just collections of norms?

Discussion Question #2

Are institutions just collections of norms?

Answer: Reasoning about when a norm becomes relevant or active in social settings requires a latent representation of the institutional structure. An account that simply enumerates injunctive and descriptive norms will be insufficient for this.

Discussion Question #2

Are institutions just collections of norms?

Answer: Reasoning about when a norm becomes relevant or active in social settings requires a latent representation of the institutional structure. An account that simply enumerates injunctive and descriptive norms will be insufficient for this.

Example: When Alex replaces Paul at the rental car agency, he inherits the service rep role's norms—not because we transferred Paul's individual norms, but because both occupy the same structural position. The institutional graph activates norms based on role occupancy, not individual identity.

Discussion Question #3

Q2) As we saw last week, Norm-augmented Markov Games (NMG) allow agents to model collective behavior as guided by shared norms:

$$P(\text{action histories}) = \sum_{\text{norms}} P(\text{norms}) \prod_i P(\text{actions of agent } i \mid \text{norms})$$

This factors into a prior over shared norms $P(\text{norms})$ and an individual, norm-independent component $P(\text{actions of agent } i \mid \text{norms})$ for each agent.

Can this generative model capture institutional representations? How would you add structure to each agent's priors over collective behavior to capture the institutional stance?

Attempt at a solution: Extended Model for Institutional Stance

NMG doesn't fully capture institutions—it assumes fixed, known roles and all agents relate to norms the same way.

Key extension: Make roles and institutional structure latent and learnable.

Original NMG: $P(\text{action histories}) = \sum_{\text{norms}} P(\text{norms}) \prod_i P(\text{actions}_i \mid \text{norms})$

Institutional model: $P(\text{action histories}) = \sum_{I,R} P(I, R) \prod_i P(\text{actions}_i \mid \text{role}_i, I, R)$

Where:

- I = institutional structure (graph of roles)
- R = role assignments for each agent
- $P(I, R) = P(I) \cdot P(R \mid I)$ factorizes into structure and assignments

Agents must now infer: (1) which institutional structure is operative, (2) which role each agent occupies, (3) internal and inter-role norms

Interplay with Mentalizing

When Do We Use Each Stance?

1) Institutional stance without mentalizing:

Reduces cognitive costs—order coffee and expect it to appear without tracking mental states, like using a vending machine.

When Do We Use Each Stance?

1) Institutional stance without mentalizing:

Reduces cognitive costs—order coffee and expect it to appear without tracking mental states, like using a vending machine.

2) Competition between stances:

When predictions fail, we switch. Coffee delayed? Start mentalizing—are they busy or did they forget? Acting unusually formal? Consider institutional context—maybe their boss is watching.

When Do We Use Each Stance?

1) Institutional stance without mentalizing:

Reduces cognitive costs—order coffee and expect it to appear without tracking mental states, like using a vending machine.

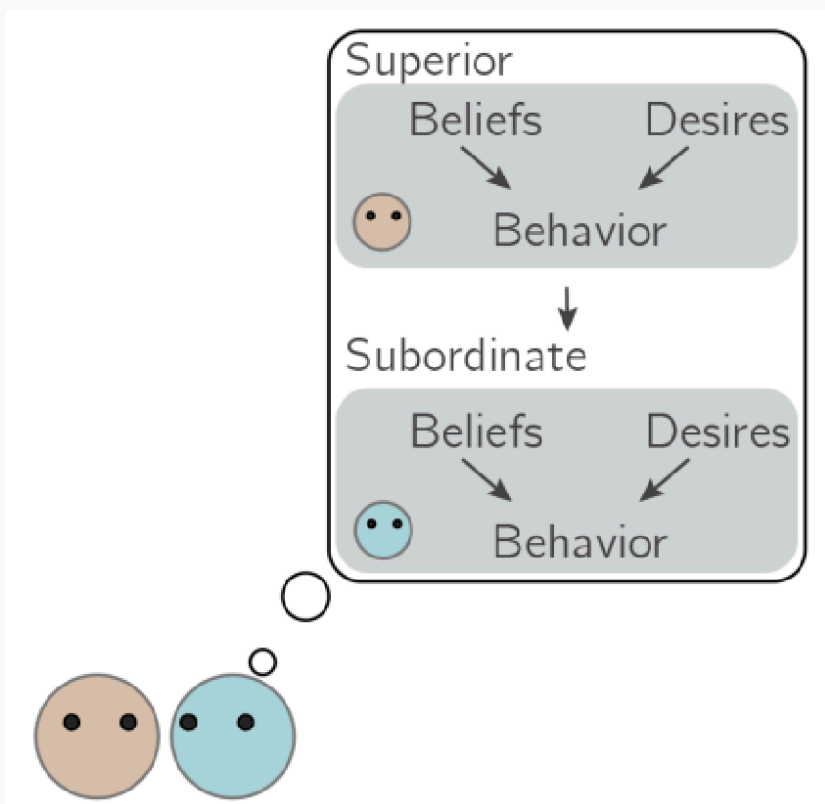
2) Competition between stances:

When predictions fail, we switch. Coffee delayed? Start mentalizing—are they busy or did they forget? Acting unusually formal? Consider institutional context—maybe their boss is watching.

3) We can also combine the stances:

Roles can encode mental state expectations, and we can attribute mental states to institutions themselves.

Institutional Stance with Mentalistic Agents

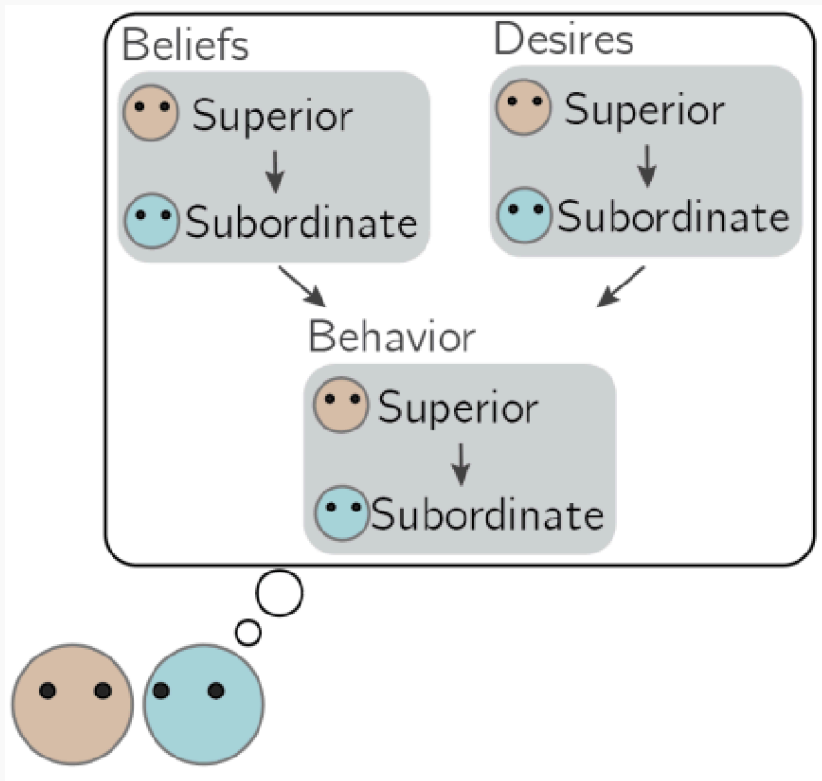


Roles can encode expectations about mental states

Example: Stock person at supermarket

- Role requires knowing where items are located
- But we don't infer they're passionate about supermarkets
- Mental states are tagged to the role, not the person

Mentalistic Stance to Understand Institutions



We attribute mental states to institutional representations themselves

Example: “Russia wants to annex Ukraine”

- Ascribing desires and intentions to groups
- Anthropomorphizing institutions and roles
- General capacity to treat collectives as mental agents

Discussion Question #4

1) Consider the way an individual agent models the group it is part of in the **Imagined We** and **Bayesian Delegation** papers.

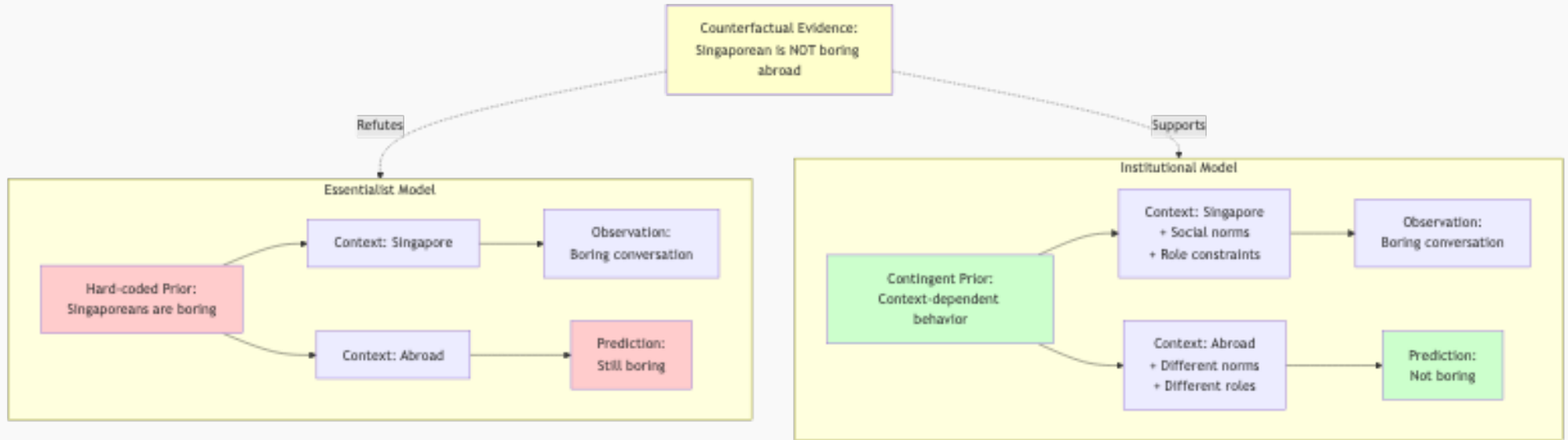
Would you describe those models as an application of the institutional stance, or the mentalistic stance? Why?

Discussion Question #5

3) The authors argue that possessing the institutional stance allows people to move beyond treating social categories like race, gender or caste as essential/immutable features, and instead understand such categories as contingent roles within institutions.

If each stance corresponds to a probabilistic generative model of the social world, how would those models differ? What evidence would cause someone to decrease belief in an essentialist model and increase belief in an institutional model?

Essentialist vs Institutional Models



Key difference: Essentialist models use hard-coded priors (stable across contexts), while institutional models use contingent priors (dependent on roles and norms).

Evidence needed: Counterfactual showing behavior changes with context proves priors are institutionally produced, not essential.

Conclusion

Implications: The Power to Create and Transform

The institutional stance enables us to:

1. **Analyze existing institutions:** Reveal hidden hierarchies and role-based systems
2. **Critique unfair systems:** Evaluate oppressive institutional structures
3. **Imagine alternatives:** Consider how to modify or erase unjust institutions
4. **Create new realities:** Synthesize novel institutional structures that genuinely create new social kinds. The generative aspect allows us to imagine and pursue genuine institutional alternatives

Questions & Further Discussion